



Accenture HackNow 2026 | Idea Submission

LOG-05 — Yard Management System (YMS) – Dwell Time and Detention Analytics

In one Sentence : A ServiceNow-powered platform for intelligent truck assignment, real-time dwell tracking, bottleneck detection, and detention cost optimization.

Team Name: NowAssen
College: K.S.R.M College Of Engineering

Team Introduction:

Team Name:

NowAssen

College Name:

K.S.R.M COLLEGE OF ENGINEERING

Team Members:

MAHAMADH BASHA VANIPENTA
GOWTHAMI KOPPOLU
GOWTHAMI GODLAVEETI
PAVAN KUMAR VAKKALAKULA

Problem Statement:

Manufacturing facilities face issues with manual gate processes, phone-based coordination, and reactive loading/unloading dock assignments. First-come-first-served decisions often ignore vehicle waiting time, priority, detention risk, and expected processing time, resulting in **long dwell times, yard congestion, underutilized docks, and increased detention costs**. Facilities also lack real-time visibility into vehicle journeys and phase-level delays, making it difficult to identify and prevent recurring operational bottlenecks.

Objective:

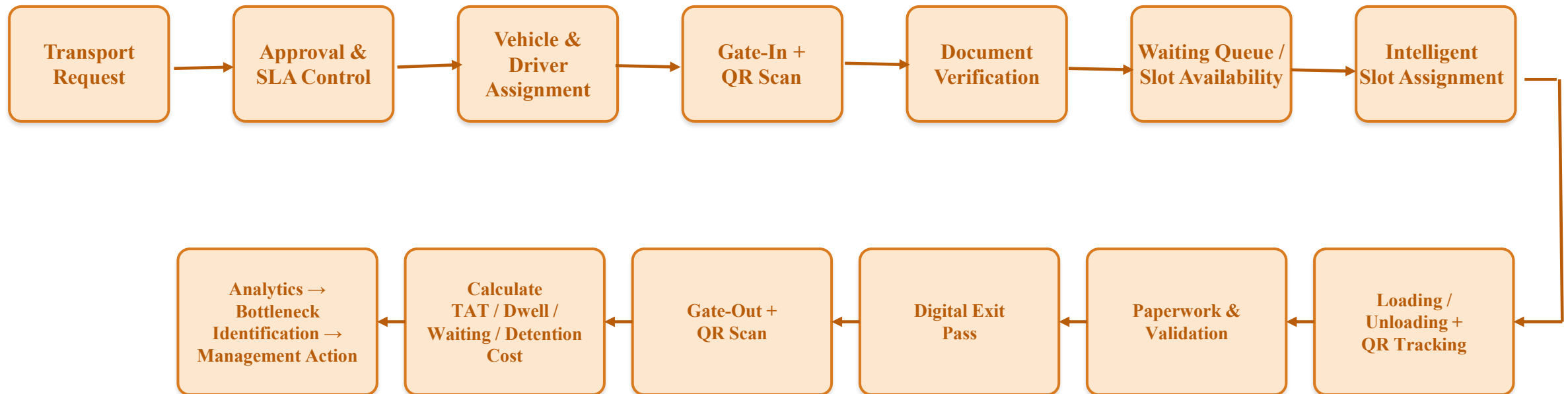
Create a **ServiceNow Intelligent Yard Management Platform** that:

- **Standardize** the complete vehicle journey from **Transport Request** → **Gate-In** → **Gate-Out**.
- Use **Vehicle Journey + Journey Phases** as the operational source of truth.
- Enable **QR-based real-time tracking** at key checkpoints.
- Use an **Explainable Intelligent Decision Engine** to assign the right vehicle to the right dock.
- Consider **waiting time, priority, detention risk, time-to-breach, and expected processing time**.
- Automatically calculate **waiting time, dwell time, turnaround time, and detention cost**.
- Identify **bottlenecks, congestion, and recurring delays**.
- Provide **real-time dashboards** for vehicle status, dock utilization, priorities, and detention costs.
- Enable **exception handling and human override** for operational control.

Approach:

We built a single ServiceNow Scoped Application centered on one core entity — Vehicle Journey — that follows every vehicle from transport request to gate-out. Each stage is captured as a Journey Phase with an expected vs. actual duration, timestamped automatically through QR scans instead of manual entry. This directly solves the core problem: yards no longer just know a vehicle was delayed — they know exactly which phase caused it, by how much, and what it cost.

Work Flow:



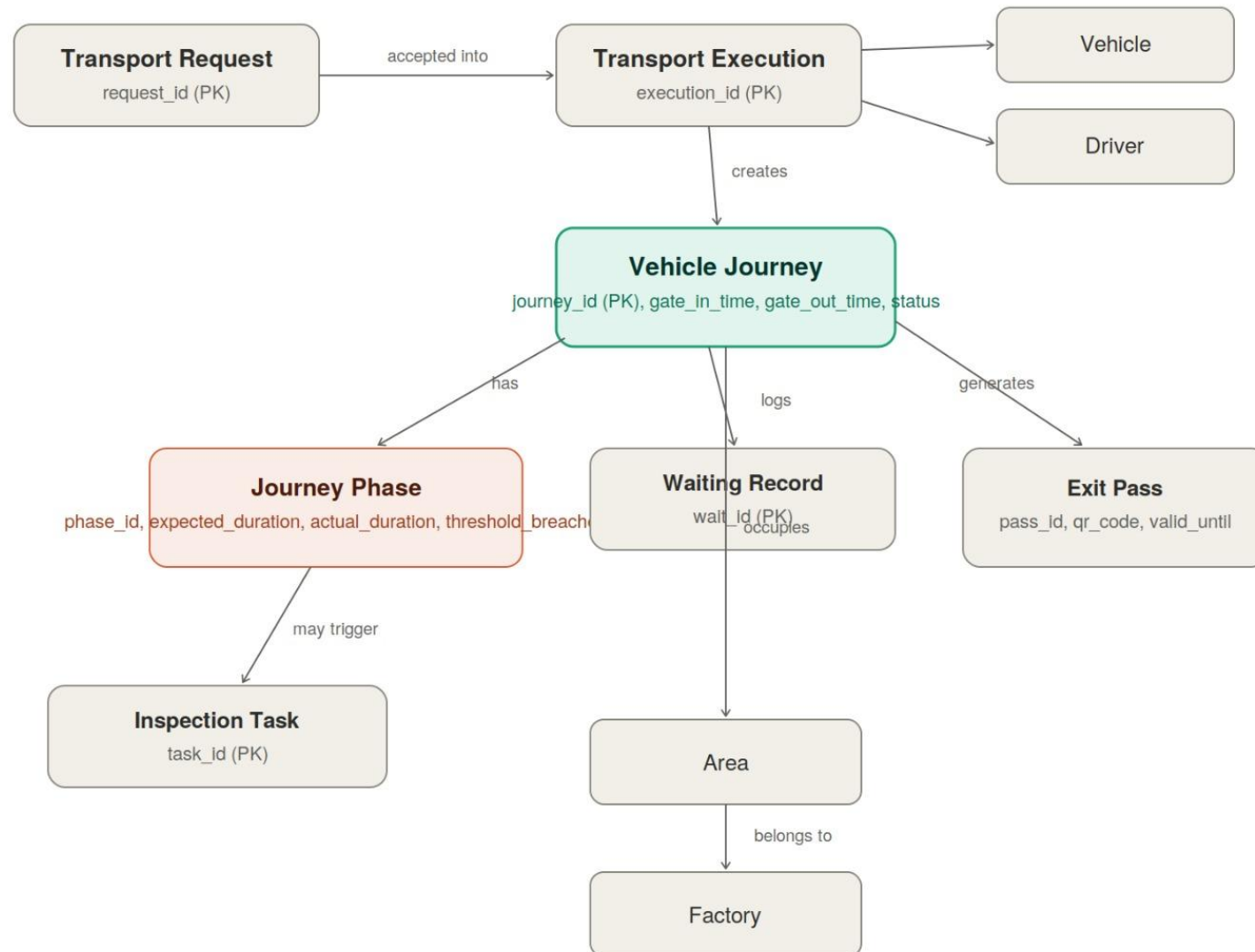
Innovation & Differentiation:

Innovation	Traditional Approach	NowAssen Approach
Intelligent dock assignment	First-come-first-served or manual choice	Score-based selection using live operational factors
Phase-level traceability	Delay entered later as free text	QR timestamps identify exact phase and lost time
Live detention cost	Calculated after the journey	Running cost visible during the journey
Explainable decisioning	Manager intuition or opaque automation	Score breakdown and decision reason logged
Human-in-the-loop	Either fully manual or fully automated	Automation recommends/assigns while authorized managers retain control
Single operational truth	Information distributed across records	Vehicle Journey + Journey Phase drive calculations and intelligence
Exception automation	Phone calls and follow-ups	Flow-based timeout, hold, inspection and notification controls
Bottleneck intelligence	Historical reporting	Phase/area/root-cause patterns drive management action

THE REAL INNOVATION

NowAssen converts yard activity into a decision loop: **Capture** → **Measure** → **Score** → **Decide** → **Explain** → **Learn**.

ER Diagram:



Why This Matters:

NowAssen turns yard delays from **unmeasured operational problems into measurable, explainable, and actionable events**.

Every vehicle's journey is tracked, every major phase is timed, slot decisions are explained, and delay impact is converted into operational and financial insight.

Track → Decide → Measure → Explain → Improve

Who is affected?

Stakeholder	Current Pain	NowAssen Benefit
Factory Transport Manager	Manual requests and status chasing	Digital request, live status and exception visibility
Transport Service Manager	Manual approvals and follow-up	Automated approval workflow and 12-hour auto-rejection
Gate Officer	Paper/manual gate processing	QR-based validation and timestamp capture
Yard / Dock Operations	Reactive slot decisions	Intelligent slot assignment and waiting-queue visibility
Drivers / Transporters	Long waiting and unclear status	Faster sequencing and clearer journey status
Factory / Transport Heads	Delayed reports	Role-based real-time dashboards
Management	Reactive decisions and unclear cost impact	Bottleneck, detention and performance intelligence

What is Unique Here

- **Vehicle Journey as Source of Truth:** One connected journey from **Transport Request** → **Gate-Out**.
- **Explainable Slot Decisioning:** Every dock assignment has a clear scoring logic and decision reason.
- **Dynamic Prioritization:** Waiting vehicles are re-evaluated whenever a dock becomes available.
- **QR-Based Phase Tracking:** Real-time timestamps are captured at major operational checkpoints.
- **Phase-Level Traceability:** Identifies exactly where time was lost and which phase caused the delay.
- **Live Detention Visibility:** Converts operational delay into a continuously updated cost view.
- **Exception & Human-in-the-Loop Control:** Managers retain control through hold, release, override, and escalation actions.
- **Bottleneck Intelligence:** Identifies recurring delays by phase, dock, area, and root cause.
- **Role-Based Decision Dashboards:** Gives factory, transport, yard, and management teams the information relevant to their decisions.
- **ServiceNow-Native Automation:** Connects workflows, approvals, QR events, notifications, analytics, and operational records in one platform.

Tools & Technologies:

ServiceNow Module	Purpose
App Engine Studio	Build the Yard Management Scoped Application
Service Portal	Portal for factory transport managers to raise and track transport requests
Flow Designer	Automate approval, gate-in, phase progression, and exit pass workflows
Integration Hub	Connect transporter systems, ERP/TMS/WMS, and future GPS/ANPR feeds
Now Mobile	App for gate officers, loading/unloading teams, and QR checkpoint scanning
Performance Analytics	Dashboards and KPIs for real-time turnaround, dwell, and detention tracking
Virtual Agent Designer	Support for drivers and transporters — request status, delay reasons, exit pass queries
Generative AI Controller (Now Assist)	AI-assisted root-cause summaries, recurring bottleneck patterns, and copilot support for yard managers
Custom Tables	Transport Requests, Vehicle Journeys, Journey Phases, Waiting Records, Exit Passes, and Inspection Tasks

Core Data Model:

Transport Request → Transport Execution → Vehicle Journey → Journey Phases → Waiting / Exit Records

Vehicle Journey acts as the **operational source of truth**, while Factory Tracking and Transport Tracking provide role-specific views.

Intelligence Layer:

- Explainable **slot-scoring engine**
- Waiting-time and priority evaluation
- Detention-risk and time-to-breach evaluation
- Bottleneck and threshold-breach analysis
- Future integration with predictive AI when sufficient historical data is available

Implementation Plan:

- **Phase 1 – Platform Setup (Week 1):**Create scoped application, custom tables, roles, Service Portal, data model.
Deliverables: Core platform & data model
- **Phase 2 – Workflow Automation (Week 2):**Build Flow Designer workflows, vehicle journey lifecycle, phase automation, notifications, dashboards.
Deliverables: Automated yard operations workflow
- **Phase 3 – AI & Integrations (Week 3):**Implement priority-based slot assignment engine, Now Assist AI skills, Integration Hub, Virtual Agent, QR-based tracking, bottleneck prediction. Deliverables: AI-powered decision support
- **Phase 4 – Testing & Deployment (Week 4):**Functional testing, UAT, performance validation, dashboard tuning, demo preparation.
Deliverables: Production-ready hackathon solution

Feasibility Assessment

- **Technical Feasibility — High:** The core solution can be implemented using **ServiceNow App Engine Studio, Flow Designer, Now Mobile, Performance Analytics, and Integration Hub** without requiring a separate platform for the MVP.
- **Operational Feasibility — High:** QR-based checkpoints provide a simple way for gate and yard teams to capture events without complex hardware. Role-based views support different operational responsibilities.
- **Implementation Feasibility — High:** The MVP can focus on the **core journey, QR tracking, workflow automation, explainable slot scoring, and dashboards**, keeping the first implementation manageable.
- **Scalability — High:** The architecture can scale from **one factory to multiple factories** by reusing the common Vehicle Journey and Journey Phase model while allowing factory-specific configurations.

Key Risks & Mitigation

Risk	Mitigation
Incorrect or missed QR scans	Validation, mandatory checkpoints and exception handling
Poor initial historical data	Start with explainable rule-based scoring
Operational resistance	Simple mobile interface and human override
Integration complexity	Keep external integrations as a future phase
Incorrect slot decision	Log score breakdown and allow manager override

Impact & Metrics

1. Measurable Outcomes

- Turnaround & Dwell Time
- Detention Cost
- On-Time Completion
- Dock Utilization
- Operational Delays
- Real-Time Visibility

Hackathon Demo Success

For the hackathon, success means demonstrating all **six KPIs updating live** as a vehicle moves through its complete journey. Judges should be able to see the impact in real time rather than viewing a static end-of-day report.

2. Real-World Applicability

The solution is designed to go beyond a single factory or hackathon demonstration. The underlying data model is built for **scalability from the beginning**.

A single **Vehicle Journey ID** can track a vehicle whether the yard has one dock or fifty. Scaling the solution therefore means adding factories, docks, and transporters rather than rebuilding the entire system.

Role Assignment

Name	Role	Responsibilities
V MAHAMADH BASHA	Team Lead / Solution Architect	Data & Platform Foundation — Architecture, data model, and judge presentation
V PAVAN KUMAR	Workflow Developer	Workflow & Automation — Flow Designer logic, approvals, and Integration Hub setup
K GOWTHAMI	AI & Logic Developer	Intelligence & AI — Scoring engine, Now Assist, and Virtual Agent
G GOWTHAMI	Experience & Analytics Developer	Experience & Analytics — Service Portal, Now Mobile, and dashboards

Leadership and Coordination

Team Leader — V MAHAMADH BASHA

V MAHAMADH BASHA is responsible for overseeing the **overall architecture, task allocation, module integration, timeline management, and final judge Q&A**. The team lead also ensures that all four technology layers work together as one complete solution.

Skill Mapping

Each team member owns **one complete technology category end-to-end**, rather than working on isolated tasks. This creates a clear owner for every major layer of the solution:

Data & Platform → Workflow & Automation → Intelligence & AI → Experience & Analytics

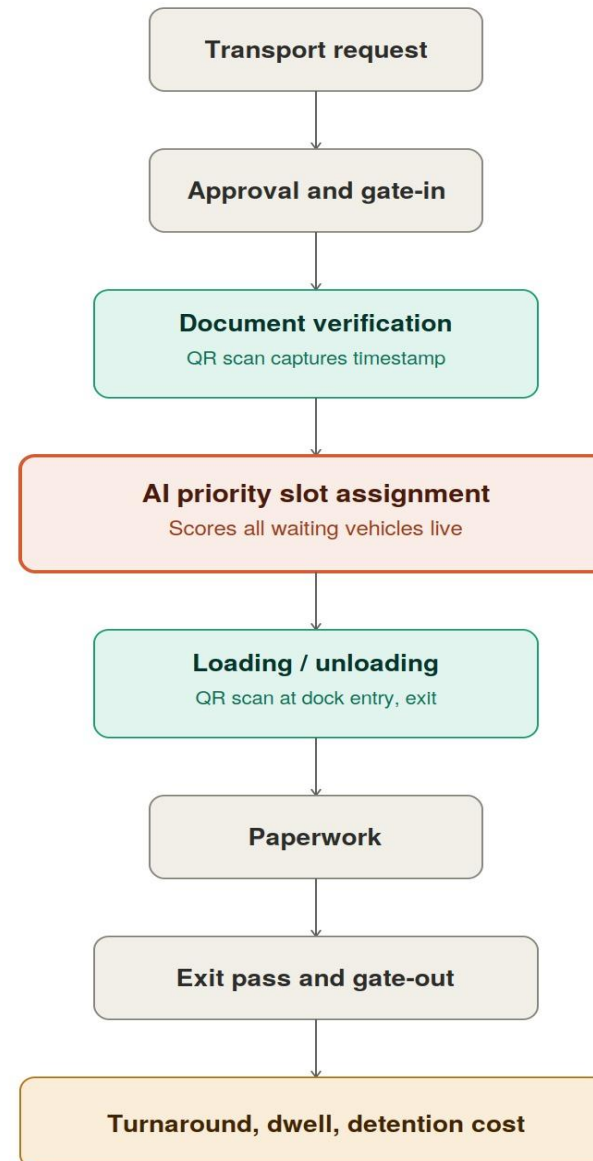
Synchronization Approach

The team will conduct **synchronization checkpoints at the end of each development week to:**

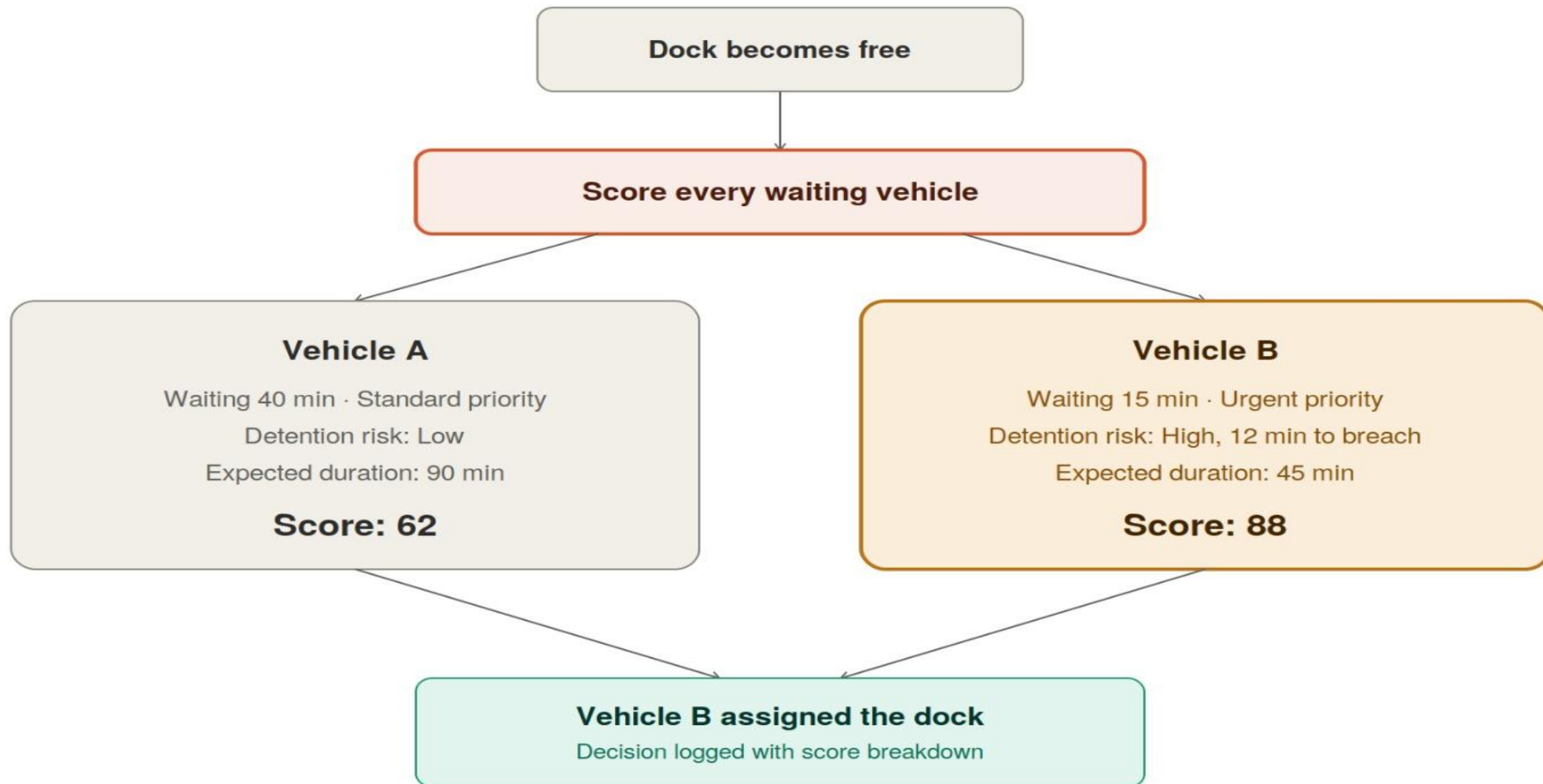
- Review progress across all modules.
- Identify and resolve integration issues early.
- Ensure dependencies between team members are completed on time.
- Validate that all modules work together as one solution.
- Keep the project aligned with the hackathon timeline and final demonstration.

Visuals Aids

1.Business Work Flow:



2.Intelligent Slot Assignment Engine: The MVP uses an explainable rule-based scoring approach. It does not require a trained machine-learning model. This keeps the hackathon implementation realistic while creating a clear path to predictive AI.

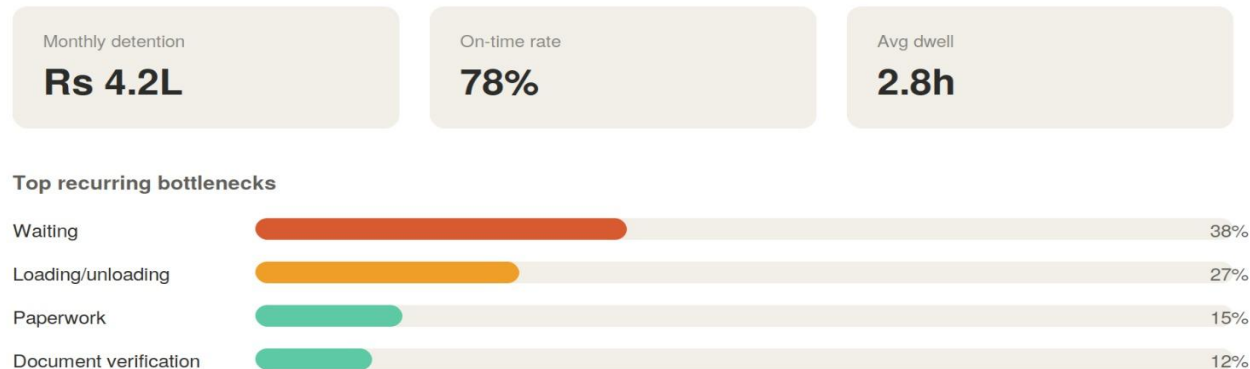


3. Real-Time Dashboards & Decision Support:

3.1 Factory Head Dashboard

- Total transport requests
- Active vehicles in factory
- Vehicles waiting
- Vehicles in loading/unloading
- Average turnaround and dwell time
- Total delay time
- Detention cost
- Area/dock performance
- Phase-wise delays and bottlenecks

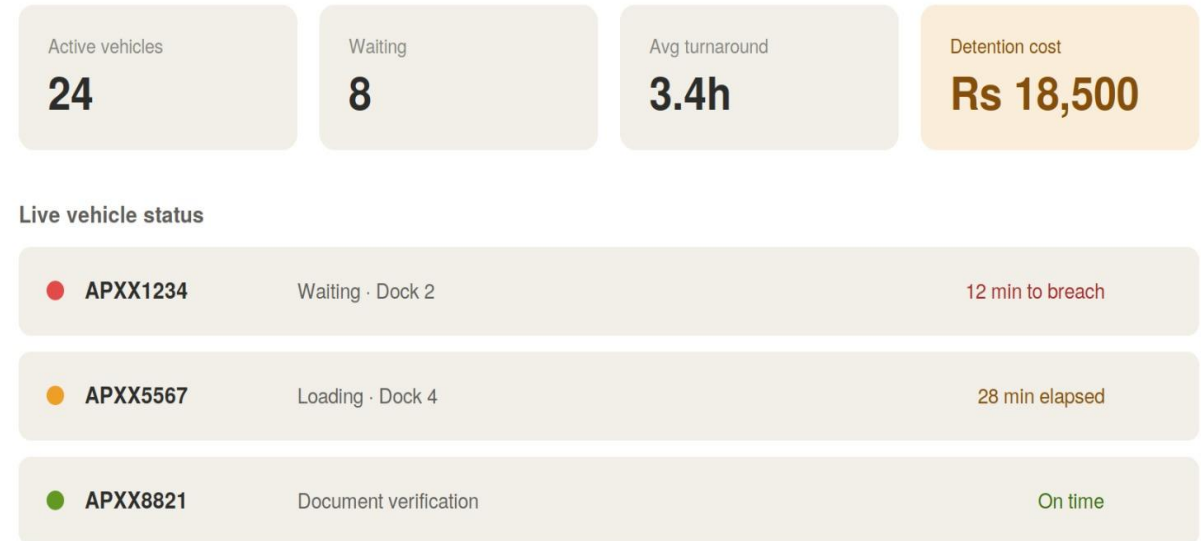
Executive dashboard



Worst performing area: Electronics loading area

42% of journeys breach threshold

Factory / Yard manager dashboard



3.2 Transport Head Dashboard

- Assigned vehicles
- Vehicles on hold
- Vehicles currently in journey
- Vehicle / transporter performance
- Waiting time and detention analysis
- Delay analysis

Future Roadmap:

Stage	Evolution
MVP	QR tracking + rule-based explainable slot scoring + real-time calculations + dashboards
Phase 1 — Pilot	One factory and one transporter; validate workflow, scoring and operational behavior
Phase 2 — Multi-Factory	Expand the same data model, KPI framework and intelligence across multiple factories
Phase 3 — Smart Technology	Integrate IoT dock sensors, ANPR gate recognition, GPS tracking and real-time ETA feeds
Phase 4 — Predictive AI	Use historical journey data for delay prediction, congestion prediction, detention-risk prediction and intelligent slot optimization

Scalability & Real-World Applicability:

Application Area	How NowAssen Helps
Manufacturing Yards	Manages Gate-In, document verification, dock assignment, loading/unloading, and Gate-Out.
Factory Logistics Operations	Prioritizes vehicles using waiting time, urgency, detention risk, and processing time.
Multi-Dock Yards	Dynamically assigns the next available dock to improve utilization and reduce waiting.
Transporter Operations	Provides visibility into vehicle journey, delays, waiting time, and detention cost.
Multi-Factory Deployment	Extends the same Vehicle Journey, workflow, KPI, and decision engine across multiple factories.

References:

- ServiceNow Developer Portal --developer.servicenow.com
- ServiceNow Product Documentation--docs.servicenow.com
- App Engine Studio Documentation--App Engine Studio
- ServiceNow App Development & Low-Code Documentation--[App Development & Low-Code](#)
- ServiceNow Mobile Platform Documentation--[Now Mobile / Mobile Platform](#)
- ServiceNow Community--[ServiceNow Community](#)
- ServiceNow Learning / Training--ServiceNow Training
- ServiceNow Platform / Product Information--ServiceNow Platform

Learning Resources :

- ServiceNowDeveloperProgram
- ServiceNow Community
- Service Now Learning Portal
- Service Now Product Documentation

*Thank
you!*